Non-Perturbative Floquet Engineering of the Toric-code Hamitonian and its ground state

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Why Error Correction?

- Avoiding randomized/lost information.

Error Sources in Quantum Computation

- Errors in gate operations.
- Crosstalk between qubits.
- Environmental deccoherence.
- Thermal noise.
- initialization errors and qubit loss
- Measurement errors.

- Helping preserving message.

Error correction in classical subsystem

Classical Repetition code

- Encode bits by repetition:

$$0_L \to 000, 1_L \to 111$$

- Codewords after a single bit flip error:

$$0_L \to 100, 0_L \to 010, 0_L \to 001$$

$$1_L \to 011, 1_L \to 101, 1_L \to 110$$

- Error detection: Bits in the string are not identical.
- Error correction: A majority vote. $010 \to 0$...

Quantum Error Correction

Barriers(Gottesman, 2009; Kitaev, 2003)

- Measurement destroys superpositions
- No-cloning theorem prevents repetition.
- Correction of multiple types of errors – e.g., bit flip and phase errors

Quantum Error Correction

Methods

- measure the error, not the data.
- Redundancy instead of repetition.
- error-correcting codes(Aharonov & Ben-Or, 1996; Shor, 1995)
  - Shor code(Devitt et al., 2013; Shor, 1995)
  - Steane code(a CSS code)(1996)
  - 5 qubit code(Knill et al., 2001)
  - All CSS codes.(1996; Calderbank & Shor, 1996)
  - Toric code(Dennis et al., 2002a; Kitaev, 2003)
  - Planar surface code(Freedman & Meyer, 1998)

Topological-protected Toric code(Kitaev, 2003)

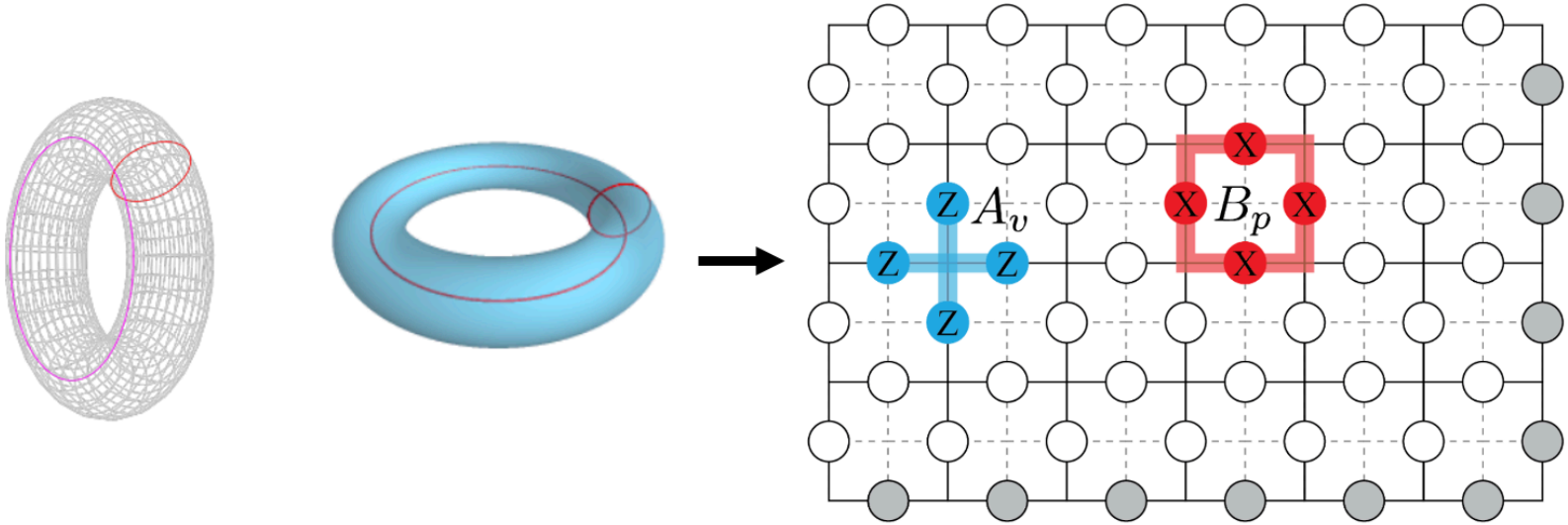

Figure 1: Left: Two loops cannot be deformed to a point or to each other. Right:square lattice with periodic boundary

$$H = -\sum_{v} A_v - \sum_{p} B_p, \left(A_v = \prod_{j \in \text{ star}(v)} Z_j, B_p = \prod_{j \in \text{ bdy}(v)} X_j \right)$$

Topological-protected Toric code(Kitaev, 2003)

- Error detection with Toric code:

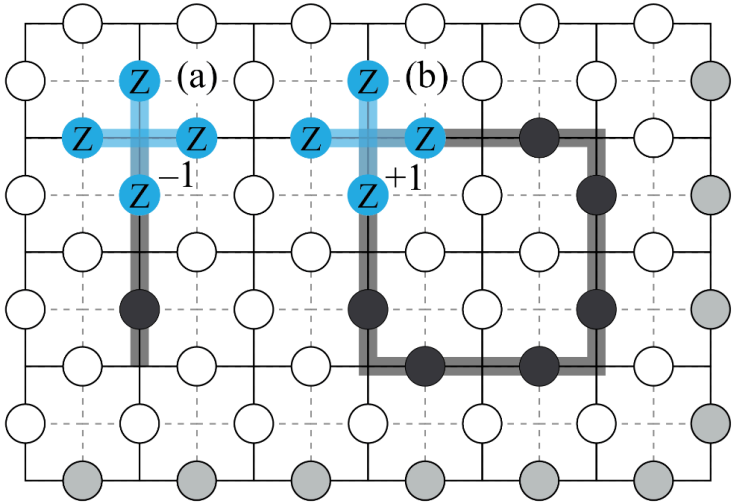

Figure 2: Error correction with Toric code.

Non-Perturbative Floquet Engineering of the Toric-code Hamitonian

Motivation

- Realizing the Toric-code Hamiltonian in 2D lattice.
  - Clean four-spin interactions.
- Methods:
  - Floquet engineering.
  - Trotterization.
- It is promising in the self-correcting machanism smaller than 4D quantum system.

Model

Hamitonians of driven/target system

- Driven system: $\hat{H}(t) = \hat{H}_1(t) + \hat{H}_2(t)$,

$$\hat{H}_1(t) = -g_{\alpha\beta} \sum_{\langle\alpha,\beta\rangle} \left(a_\alpha^\dagger a_\beta + a_\beta^\dagger a_\alpha\right) \to \sum_{\langle\alpha,\beta\rangle} g_{\alpha\beta}\left(\hat{X}_\alpha \hat{X}_\beta + \hat{Y}_\alpha \hat{Y}_\beta\right)$$

- Target system: Wen's plaquette model(Brown et al., 2011)(=Kitaev's model)(Nussinov & Ortiz, 2009) on a square lattice

$$\hat{H}_\omega = -J \sum_{i,j} \hat{P}_{i,j}$$

$$\hat{P}_{i,j} = \hat{X}_{i,j} \hat{Z}_{i,j+1} \hat{Z}_{i+1,j} \hat{X}_{i+1,j+1}$$

Why simulating Wen's Model

- Wen's formulation:

  – a promising surface code dessign against biased noise(Acharya et al., 2022; Bonilla Ataides et al., n.d.).

- Method: Finding control functions $g_{\alpha\beta}$ and Ω_α.

Floquet engineering four-spin interactions

Central difficulty

- clean four-spin interactions.

Floquet engineering

- Considering a foundamental four-spin plaquette.
- For a time step τ, effective Hamiltonian $H_{i,j}$ at τ is derived through Magnus expansion.

$$\exp\left(-i\hat{H}_{i,j}\tau\right) = \tau \exp\left(-i\int_0^\tau \mathrm{d}t\hat{H}(t)\right)$$

- Search for periodic $g_{\alpha\beta}(t) \to H_{i,j} \approx -JP_{i,j}$.

Single plaquette model

Single plaquette model

- Strong resonant X-drive: $\hat{H}_2 = \Omega_1 \hat{X}_1 + \Omega_4 \hat{X}_4, (\Omega_{1,4} \gg g_{\mathrm{kk'}(t)})$.

(Penalizing the process induced by $\hat{Y}\hat{Y}$ coupling.)

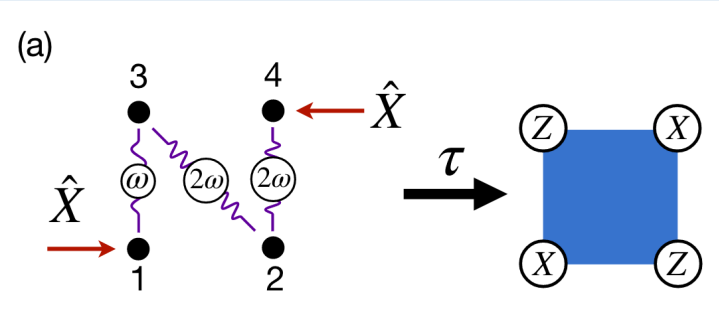

Figure 3: Sketch of the driving scheme for realizing a fourspin term in a single four-spin plaquette.

- $\hat{H}(t) = \Omega_1 \hat{X}_1 + \Omega_4 \hat{X}_4 + \sum_{\langle k,k' \rangle} g_{\mathrm{kk'}} \left(\hat{X}_k \hat{X}_{k'} + \hat{Y}_k \hat{Y}_{k'} \right)$

Single plaquette model

Driving functions

$$g_{13} = g_{13}\cos(\omega t)$$

$$g_{23} = g_{13}\cos(2\omega t)$$

$$g_{24} = g_{13}\cos(2\omega t)$$

$$\omega = 2\pi/\tau$$

- It is time symmetric : $g_{\mathrm{kk'}}(t) = g_{\mathrm{kk'}}(\tau - t)$

Single plaquette model

Effective Hamitonian

- All possible operators ← dynamical Lie algebra of the control system.
- Effective Hamiltonian

$$\hat{H}_{i,j} = \sum_{O \in L} c_l \hat{O}_l$$

- Challenge: Single out the desired term.

single plaquette model

Strong and clean 4-body-interaction

- Method: Optimize $g_{\mathrm{kk'}}, \Omega_j$ via exact numerical simulation in single-plaquette model rather than high-frequency expansion.
- The four-body interaction is the dominant term:

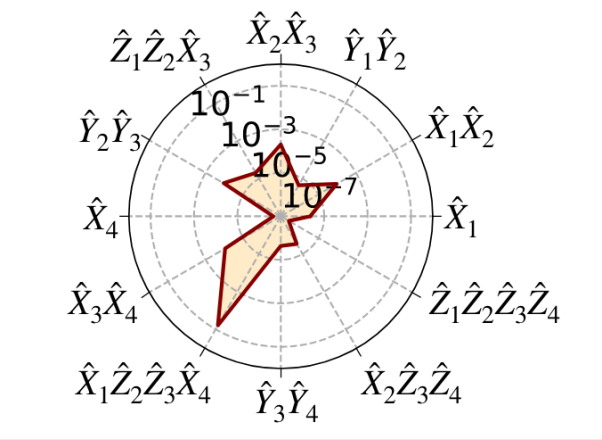

Figure 4: Structure of the lattice.

single plaquette model

Effective Hamiltonian

- Effective Hamiltonian

$$\hat{H}_{i,j} = -J\hat{P}_{i,j} + \varepsilon\hat{V}_{i,j}$$

  The extra terms are contained in $\hat{V}_{\mathrm{i,j}}$ and $\varepsilon/J = 1.5 \cdot 10^{-3}$.
- Dissipation process: The decoherence and dissipation rates are not significantly impacted by the Floquet driving process.(Appendix C)

single plaquette model

Ground state

- Toric ground state for mixed boundary conditions:

$$|G\rangle = \frac{1}{2}\left(1 + \hat{Z}_1\hat{X}_2\right)(1 + \hat{X}_3\hat{Z}_4|-11-\rangle$$

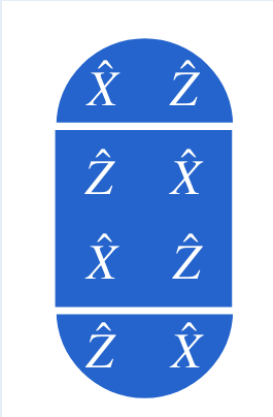

Figure 5: Two-by-two toric-code Hamiltonian with mixed boundary conditions.

Full lattice sequence

first-order Floquet-Trotter sequence

- Devide the Hamitonian as the sum of four parts:

$$\hat{H}_k = \sum_{(i,j)\in G_k} \hat{H}_{i,j}$$

- Consider one substep τ rather than the whole period T.
- The propogator:

$$\hat{U}(T) = \prod_{k=1}^{4} \prod_{(i,j)\in G_k} \hat{U}_{\mathrm{ij}(\tau)}$$

Full lattice sequence

where $U_{\mathrm{ij}}(\tau) = e^{-i\tau\hat{H}_{i,j}}$ The product of propogators $\hat{U}_k = e^{-i\hat{H}_k\tau}$ belonging to two different groups do not commute and can be estimated by expontial product formula.

$$\prod_k \hat{U}_k = \exp\left(-i\tau\sum_k \hat{H}_k + \frac{\tau^2}{2}\sum_{j<k}\left[\hat{H}_j, \hat{H}_k\right] + \ldots\right)$$

- In effective Hamitonian, the term $P_{i,j}$ commutes with each other, so the equation above is used to calculate error terms.

$$\hat{H}_{\mathrm{eff}} = \hat{H}_\omega + \widehat{\mathrm{Err}}$$

Full lattice sequence

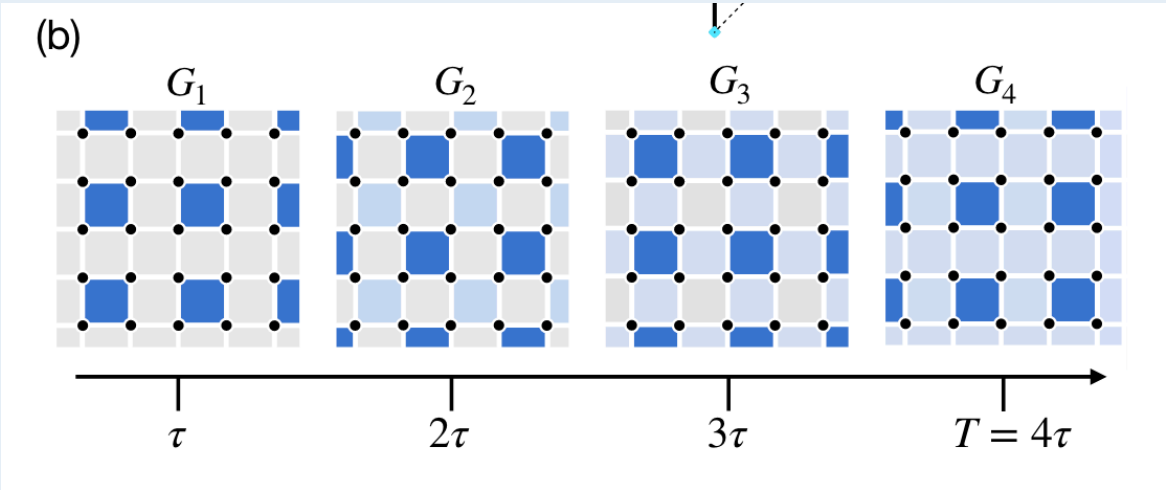

Figure 6: Trotter sequence to realize the Hamiltonian on the whole lattice.

Simulation in 5x5 lattice

- The boundary terms involving $\hat{X}_\alpha \hat{Z}_\beta$ terms are Floquet engineering with driving scheme:

$$g_{\alpha\beta}\cos(\omega t)\left(\hat{X}_\alpha \hat{X}_\beta + \hat{Y}_\alpha \hat{Y}_\beta\right) + \tilde{g}\sin(\omega t)\hat{Y}_\beta$$

- Mixed boundary condition.

Simulation in 5x5 lattice

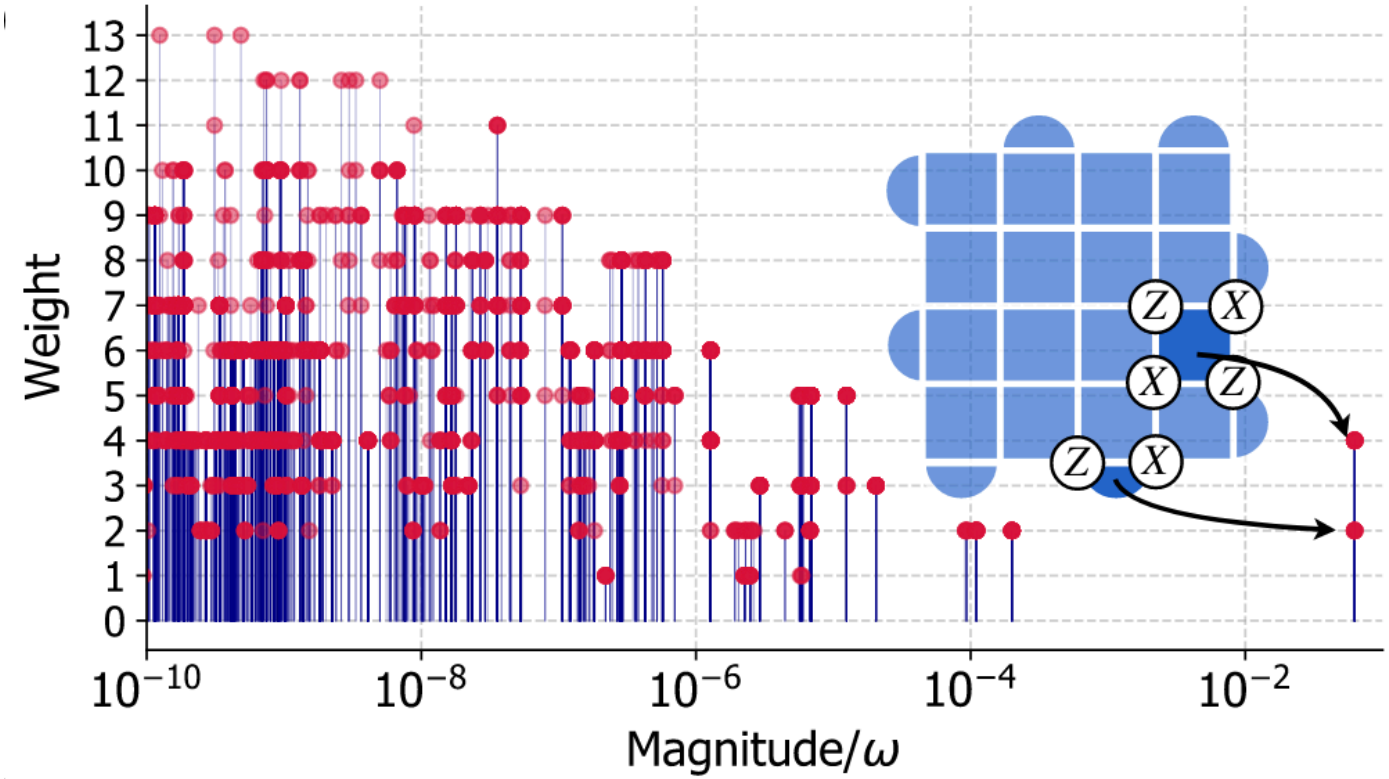

Figure 7: 5x5 lattice with mixed boundary conditions.

Present of entanglement

- Calculation of topological entanglement entropy S_{topo} with three subsystems A,B,C.
- $S_{\text{topo}} = S_A + S_B + S_C + S_{\text{AB}} + S_{\text{BC}} + S_{\text{AC}} + S_{\text{ABC}}$, where $S_X = \text{tr}\,\rho_X \log \rho_X$ is von Neumann entropy.
- $S_{\text{topo}} = -\log 2$ is very close to the value of ideal toric code, characterizing the Z_2topological order.

Why Self-correcting in Quantum computation?

- No-cloning theorem.
- Self-correction.
- Crucial mechanism(Brown et al., 2016):
  - topological protection.
  - energy barriers.
  - energy cost for local error in system.

Self-correcting in classical system

2D Ising model(Brell, 2016; Peierls, 1936)

$$H = -\sum_{i,j} Z_i Z_j$$

– 2-fold degenerate, Extensive distance, Finite temperature ordered phase, Expontential memory lifetime.

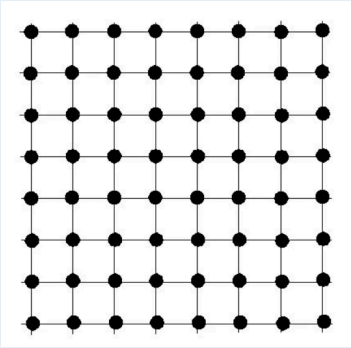

Figure 8: 2D Ising model.

Self-correcting in quantum system

- fluctuations(Alicki et al., 2009; Castelnovo & Chamon, 2007; Nussinov & Ortiz, 2008):
  - external thermal fluctuations.
  - quantum fluctuations.
- Non-local entanglement.
- Method:
  - topological order.
  - energy gap.

Self-correcting Quantum Memory

Core Requirements: The Caltech rules(Brell, 2016; Landon-Cardinal et al., 2015)

- Expontential lifetime.
- finite spins and bounded local interactions
- perturbative stability
- nontrivial codespace
- efficient decoding

Self-correcting under 2D Quantum system

2D No-Go theorem

- A self-correcting quantum memory cannot be built using stabilizer codes in 1D/2D(Bravyi & Terhal, 2009).
- Spetial operation in 2D:

– Toric-boson model(Hamma et al., 2009)

– entropic barriers(Brown et al., 2014)

Self-correcting under 3D Quantum system

3D No-Go theorem

- No local spin models in 2D or 3D are known to be fully self-correcting. But many approaches towards realizing some aspects of self-correction in 3D have also been found

– Haah code(Lin et al., 2024),(Bravyi & Haah, 2013; Haah, 2011): 3D cubit code, self-correction in 3D cubic model.

– Welded codes: Combining two stabilizer codes for a new stabilizer code and then use it to construct solid codes.(Michnicki, 2012)

– subsystem codes:encode information into a subsystem of a Hilbert space.(Bacon, 2006)

- bosonic bat model: Locally coupling a 2D code to a 3D bath of bosons hopping on a cubic lattice.(Pedrocchi et al., 2013)

Self-correcting under 3D Quantum system

- Four spatial dimensions can achieve self-correctihg(4D Toric code(Dennis et al., 2002b), (Alicki et al., 2010)).

Self-correcting under 4D Quantum system

4D Self-correcting theorem

- Four spatial dimensions can achieve self-correctihg: 4D Toric code(An example for 4D Caltech SCQM). (Alicki et al., 2008),(Dennis et al., 2002b),(Alicki et al., 2010).

$$H = -\sum_{l} A_l - \sum_{c} B_c \left(A_i = \prod_{j \in i} X_j, b_k = \prod_{j \in k} Z_j \right)$$

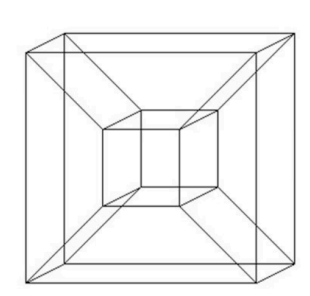

Figure 9: 4D Toric code.

Past Works

- A proposal for self-correcting stabilizer quantum memory in 3D(or slightly less)(Brell, 2016)

Classical system	Quantum system
1D Ising model	2D Toric code
2D Ising model	4D Toric code
???	3D quantum system

- Inspiration:

– classical Ising model on the Sierpinski carpet fractal.

- Quantum-system dimension:

– on fractal subsets of a 4D hypercubic lattice with Hausdorff dimension less than 3.

Past Works

Figure 10: Sierpinski carpet fractal. Its dimension is less than 2.

Past Works

- Symmertic-protected Self-Correcting Quantum Memories(Roberts & Bartlett, 2020)

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